Real-World Applications & Global Change Projections

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Conservation & Management

2 min. PITCH

Papers: P21, P24, P30

10 minutes

Read the abstract and the introduction and interpret the main figures.

Question. What problem did the authors tackle? (one sentence)

Approach. What did they do, and with which data and model? (one sentence)

Key finding. The one result to remember, ideally with the paper's main figure.

Why it matters today?

How the papers contribute to conservation and management?

Identifying and Protecting Vulnerable Habitats

Contents lists available at ScienceDirect

Marine Environmental Research

journal homepage: http://www.elsevier.com/locate/marenvrev



Environmental drivers of rhodolith beds and epiphytes community along the South Western Atlantic coast

Vanessa F. Carvalho [a,*], Jorge Assis [b], Ester A. Serrão [b], José M. Nunes [c], Antônio B. Anderson [d], Manuela B. Batista [a], José B. Barufi [a], João Silva [b], Sonia M.B. Pereira [e], Paulo A. Horta [a,f,g,**]

[a] Laboratório de Ficologia, Departamento de Botânica, Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina, Brazil
[b] CCMAR – Centre of Marine Sciences, Universidade do Algarve, Campus de Gambelas, Faro, Portugal
[c] Instituto de Biologia, Universidade Federal da Bahia, Salvador, Bahia, Brazil
[d] Universidade Federal do Espírito Santo - Programa de Pós-graduação em Oceanografia - Laboratório de Ictiologia (Ictiolab) - Campus Goiabeiras - Vitória - ES - Brazil
[e] Departamento de Biologia, Universidade Federal Rural de Pernambuco, Recife, Brazil
[f] Programa de Pós Graduação em Oceanografia, Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina, Brazil
[g] Programa de Pós Graduação em Ecologia, Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina, Brazil

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ABSTRACT

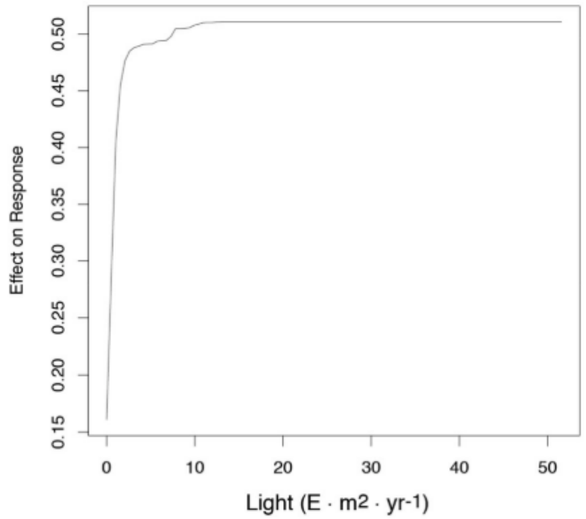
Environmental conditions shape the occurrence and abundance of habitat-building organisms at global scales. Rhodolith beds structure important hard substrate habitats for a large number of marine benthic organisms. These organisms can benefit local biodiversity levels, but also compete with rhodoliths for essential resources. Therefore, understanding the factors shaping the distribution of rhodoliths and their associated communities along entire distributional ranges is of much relevance for conservational biology, particularly in the scope of future environmental changes. Here we predict suitable habitat areas and identify the main environmental drivers of rhodoliths' variability and of associated epiphytes along a large-scale latitudinal gradient. Occurrence and abundance data were collected throughout the South-western Atlantic coast (SWA) and modelled against high resolution environmental predictors extracted from Bio-Oracle. The main drivers for rhodolith occurrence

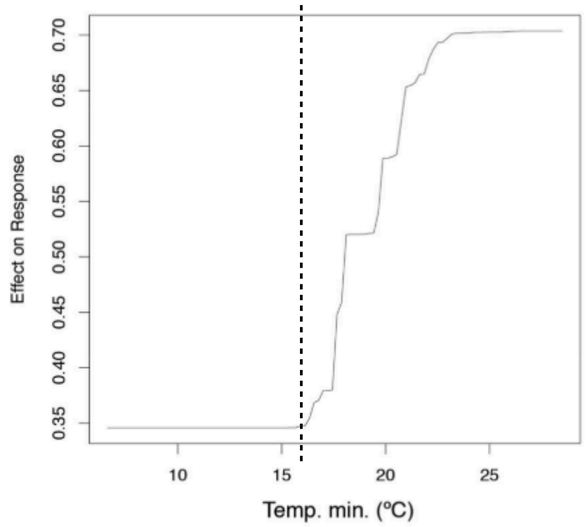


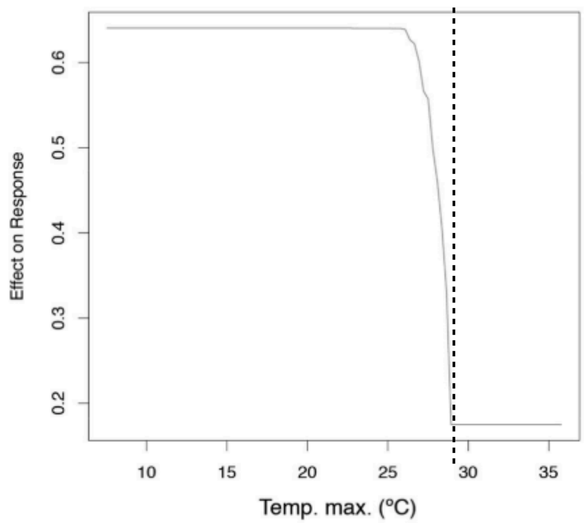
Rationale: Rhodolith beds structure important habitats for a large number of organisms. However, their **deep and cryptic occurrence hinders knowledge about where they are, and about the factors shaping their distribution**.

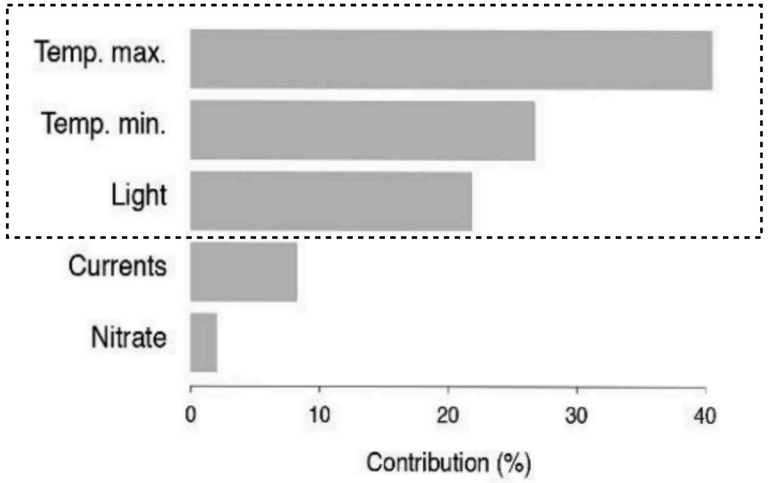
Goal: identify the distribution drivers and patterns of rhodoliths.

Methods: Occurrence records collected throughout the South-western Atlantic coast and modeled against environmental predictors extracted from Bio-Oracle.

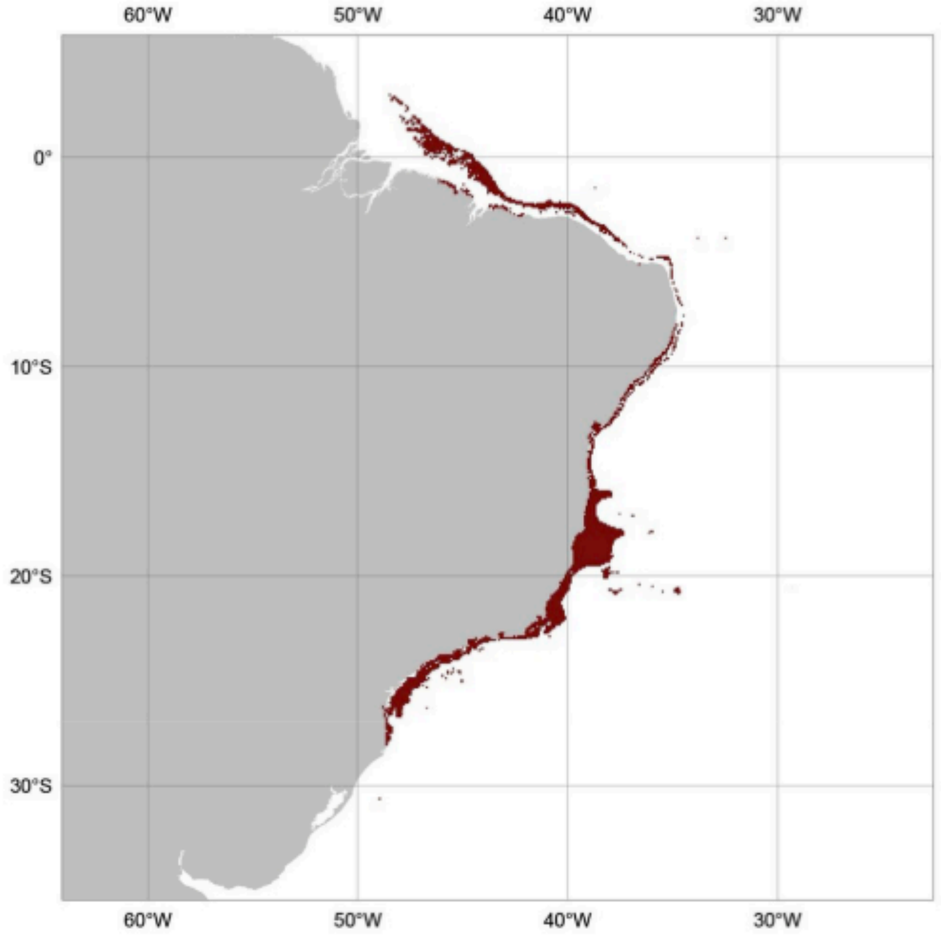








Results: main drivers for rhodolith distribution are light availability and temperature, with clear **ecological responses supported by additional empirical studies**.



Results: The first map of potential distribution of rhodolith beds. Brazilian tropical-temperate region provides broad suitable areas, decreasing toward extreme warm and cold regions.



LETTERS

Brazil oil spill response: Protect rhodolith beds

In his News In Depth story "Mystery oil spill threatens marine sanctuary in Brazil" (8 November 2019, p. 672), H. Escobar highlights important ecosystems that have been affected by the spill. However, he did not mention the Brazilian rhodolith beds—the most extensive, abundant, and diverse biogenic carbonate habitats in the South Atlantic (*1*). The oil spill severely threatens these ecosystems, which comprise a staggering 2 x 10^{11} tons of carbonatic bank (*2*), stretch from 5°N to 27°S along the Brazilian coast, and cover a seabed potential area of 229,000 km^2 (*1*).

Brazil's rhodolith beds are recognized as an oasis of diversity (*3*). Although they harbor species of great economic and ecological value, they remain unprotected. The oil pollution will likely cause major socio-environmental and economic losses,

Identifying and Protecting Vulnerable Habitats



frontiers in Marine Science

ORIGINAL RESEARCH
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Check for updates

Bottom Trawling Threatens Future Climate Refugia of Rhodoliths Globally

Eliza Fragkopoulou[1], Ester A. Serrão[1], Paulo A. Horta[2], Gabrielle Koerich[2] and Jorge Assis[1]*

[1] *Center of Marine Sciences, University of Algarve, Faro, Portugal,* [2] *Phycology Laboratory, Department of Botany, Biological Sciences Center, Federal University of Santa Catarina, Florianopolis, Brazil*

Climate driven range shifts are driving the redistribution of marine species and threatening the functioning and stability of marine ecosystems. For species that are the structural basis of marine ecosystems, such effects can be magnified into drastic loss of ecosystem functioning and resilience. Rhodoliths are unattached calcareous

Evaluate new areas for invasive species

Science of the Total Environment 700 (2020) 134692

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Science of the Total Environment

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How experimental physiology and ecological niche modelling can inform the management of marine bioinvasions?

Gabrielle Koerich [a,c,*], Jorge Assis [b], Giulia Burle Costa [a,c], Marina Nasri Sissini [a,c], Ester A. Serrão [b], Leonardo Rubi Rörig [a], Jason M. Hall-Spencer [d], José Bonomi Barufi [a], Paulo Antunes Horta [a,*]

[a] Phycology Laboratory, Botanical Department, Federal University of Santa Catarina, 88040-970 Florianópolis, Santa Catarina, Brazil
[b] Centre of Marine Sciences, CCMAR-CIMAR, University of Algarve, Campus Gambelas, 8005-139 Faro, Portugal
[c] Post-graduate Program in Ecology, Federal University of Santa Catarina, 88040-970 Florianópolis, Santa Catarina, Brazil
[d] Marine Biology and Ecology Research Centre, Plymouth University, Drake Circus, Plymouth PL4 8AA, United Kingdom

HIGHLIGHTS

- An Ecological Niche Model was built to explore the distribution of an invasive alga.
- Model results were crosschecked with ecophysiological evaluations.
- *G. turuturu* has higher habitat

GRAPHICAL ABSTRACT

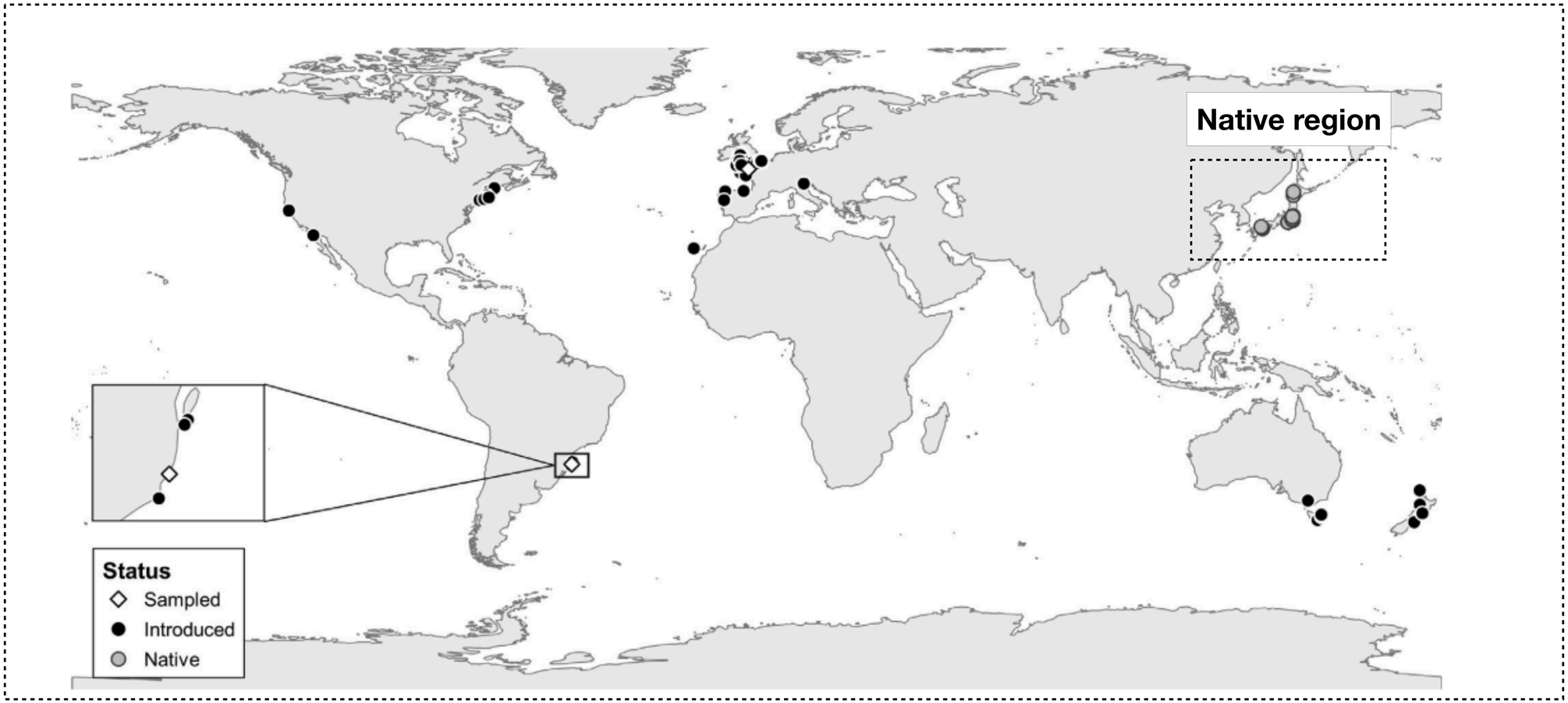


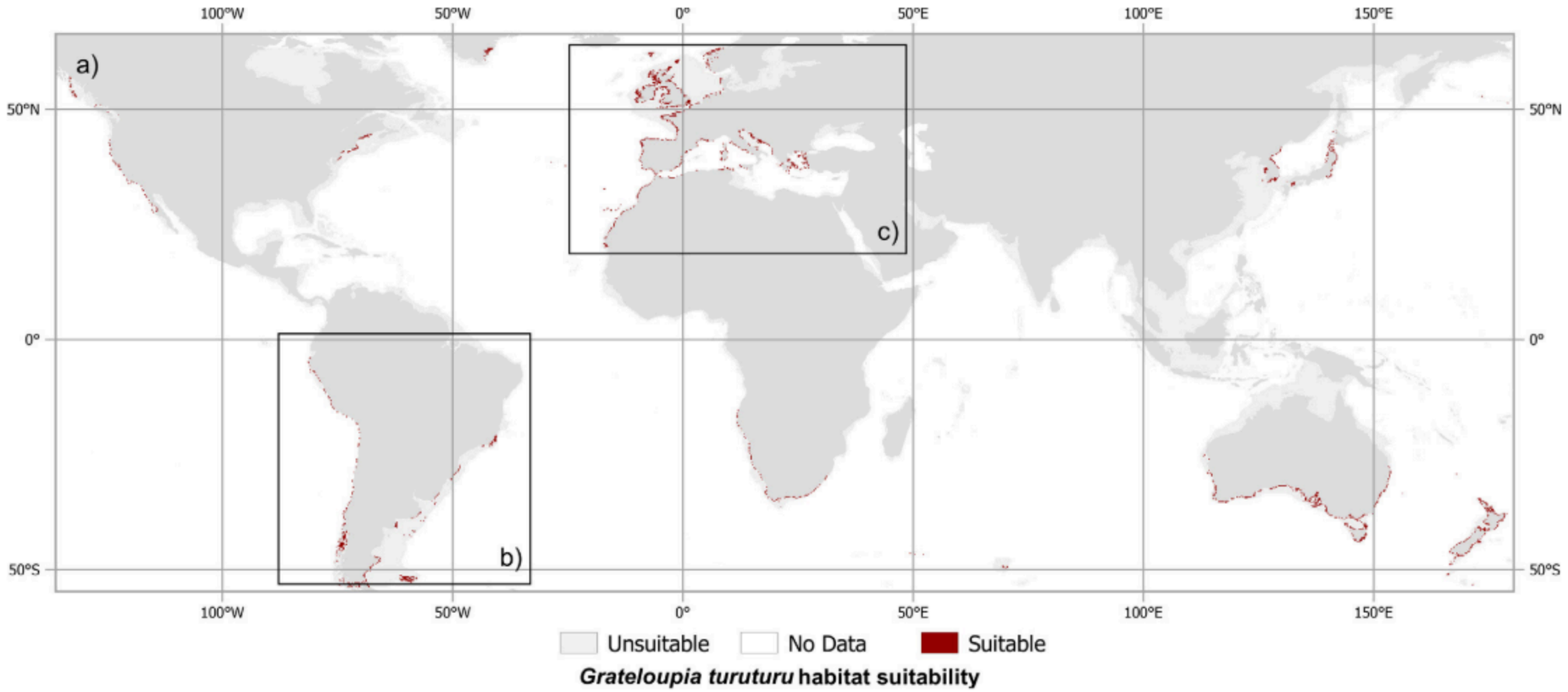


Rationale: information on **habitats of species with high invasive potential is crucial for management decisions** aiming to prevent their arrival and spread. *Grateloupia turuturu*, is one of the most harmful species, capable of causing biodiversity losses.

Goal: infer the potential global distribution to **assess potential areas of introductions**.

Methods: records collected by field sampling and the available literature, and modeled with high resolution environmental predictors extracted from Bio-Oracle.


Native region
Status
Sampled
Introduced
Native



Results: **high suitability in temperate and warm regions** around the world, with **many areas where this species still doesn't occur**. Management initiatives can mitigate anthropogenic transport and promote eradication, particularly **in the South America, South Africa, and Western Australia**.